

SUMMARY

Machine Learning Engineer with nearly **4 years of experience** designing, deploying, and scaling predictive models across automotive, retail, and fintech domains. Proficient in Python, SQL, TensorFlow, PyTorch, Scikit-learn, and cloud platforms (AWS, Azure). Achieved measurable results including 23% improvement in predictive maintenance accuracy, 17% uplift in marketing ROI, and 40% reduction in retraining cycles through advanced NLP (BERT, GPT), deep learning (CNN, LSTM), and MLOps practices (Airflow, MLflow, Docker, FastAPI). Skilled at delivering explainable, compliant AI solutions with SHAP and LIME, driving cost savings, operational efficiency, and enterprise-wide adoption.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Scikit-learn), SQL, Git/GitHub, Data Cleaning & Preprocessing
Machine Learning: Supervised Learning (Regression, Classification), Unsupervised Learning (Clustering: K-Means, DBSCAN), Model Evaluation Metrics (Accuracy, Precision, Recall, F1, ROC-AUC)
Deep Learning: CNN (Images), RNN/LSTM (Time-Series), Transformers (NLP/Text)
NLP & Generative AI: Hugging Face Transformers, BERT/GPT Models, Large Language Models (LLMs)
Deployment & MLOps: Flask / FastAPI (REST API), Docker, MLflow / Weights & Biases, Airflow / Prefect (pipelines), AWS SageMaker / EC2 / S3
Cloud & Edge Deployment: AWS / GCP / Azure basics, TensorFlow Lite, ONNX, Serverless (Lambda)
Data Visualization & Reporting: Matplotlib, Seaborn, Power BI, Tableau
Mathematics & Statistics: Probability, Linear Algebra, Statistics, Optimization (Gradient Descent), Hypothesis Testing
AI Explainability & Ethics: SHAP, LIME, Bias Detection

PROFESSIONAL EXPERIENCE

General Motors, AL, USA Machine Learning Engineer	Feb 2025 - Current
- Improved predictive maintenance accuracy 23% by deploying regression and classification models in AWS SageMaker, reducing unplanned downtime and warranty claims worth \$4.5M annually. - Increased sentiment classification precision to 91% using BERT NLP pipelines, cutting average customer service resolution time 18% across 1.2M calls. - Reduced false positives 28% by optimizing CNN-based defect detection models, accelerating visual inspection for 12 production facilities. - Achieved 40% faster retraining cycles by orchestrating automated ML pipelines with Airflow and experiment tracking in MLflow. - Delivered sub-200ms latency on 1M+ daily API requests by containerizing inference services with Docker and deploying via FastAPI. - Saved \$2M in quality control costs by enabling real-time defect detection at the edge using TensorFlow Lite models. - Reduced manual compliance review hours 60% by integrating SHAP/LIME explainability, streamlining safety audit approvals. - Boosted operational reliability 35% through monitoring drift and automating model health checks across deployed production pipelines. - Increased cross-team adoption of ML models by 45% through integrating outputs directly into BI tools used by plant managers.	
Cybage Software, India Machine Learning Engineer	May 2020 - May 2023
- Raised targeted campaign ROI 17% by delivering K-Means/DBSCAN customer segmentation models, generating \$1.2M incremental revenue. - Improved demand forecast accuracy 19% with LSTM time-series models, lowering inventory overstock 22% and reducing carrying costs. - Reduced deployment timelines 35% by building Flask APIs on AWS EC2, accelerating delivery of ML features for client dashboards. - Cut reporting workload 45% by developing executive dashboards in Tableau and Power BI, enabling faster leadership decisions. - Achieved regulatory reporting compliance by validating ML outputs using ROC-AUC, F1, and precision-recall metrics. - Reduced drift incidents 30% by implementing continuous model monitoring and retraining triggers. - Accelerated data preprocessing 25% through optimized Pandas/Numpy pipelines, improving training time for high-volume datasets. - Boosted user engagement 12% by integrating ML insights into client CRM platforms, enhancing personalization for financial services customers. - Lowered infrastructure costs 20% by fine-tuning compute resources on AWS for model training workloads.	

EDUCATION

Masters in Computer Science Auburn University at Montgomery, Montgomery, AL, USA	Dec 2024
Bachelors in Computer Science SR Engineering College, Telangana, India	Aug 2021

PROJECTS

AI-Powered Cyber Threat Detection System
- Designed and deployed intrusion detection models (Random Forest, Gradient Boosting, LSTM Autoencoders) on CICIDS2017 dataset, achieving 94% F1-score and reducing false alarms by 28%. - Built a real-time detection pipeline using Flask, Docker, AWS Lambda, enabling sub-second response times and explainable threat insights with SHAP.
BotVerse – AI-Driven Conversational Chatbot
- Developed an NLP chatbot with BERT embeddings and TensorFlow/Keras, boosting intent classification accuracy to 92% and increasing user satisfaction by 85%. - Deployed as a cloud-native service with FastAPI, Docker, AWS EC2, integrating MLflow tracking to cut retraining time by 30%.